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Penetrating Chest Trauma With Lung Evisceration, Pneumomediastinum, Pneumocranium, and Brachial Plexus Injury: A Case Report From a Resource-Limited Setting

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ABSTRACT

High-energy open chest injuries with lung evisceration may be associated with concurrent injuries and serious potential sequelae including pneumocranium, brachial plexus injury, and venous thrombosis. Early resuscitation, adherence to ATLS principles, and coordinated multidisciplinary care are critical for survival and functional outcomes, particularly in resource-limited trauma systems.

1 | Introduction

Chest trauma is a major cause of injury-related morbidity and mortality worldwide, accounting for approximately 10% of trauma admissions and up to 25% of trauma-related deaths [1]. In sub-Saharan Africa, it remains a common surgical emergency and a significant contributor to preventable trauma mortality, particularly in patients with multiple injuries [2].

Chest trauma can be broadly classified as blunt or penetrating [3]. Blunt trauma, often from road traffic collisions, falls, or industrial accidents, causes rib fractures, pulmonary contusions, and shear injury to intrathoracic viscera, sometimes involving the heart and great vessels [4, 5]. Penetrating trauma, typically from stab wounds, gunshots, or industrial machinery, carries higher immediate mortality, especially near the heart or pericardium, and may result in open chest wounds, lung evisceration, and massive hemothorax [3].

Outcomes following chest trauma are largely determined by associated injuries and physiological derangement at presentation [6]. Predictors of mortality include advanced age, comorbid conditions such as cardiac or respiratory disease, preinjury medication use, and abnormal vital signs, including tachycardia, tachypnea, hypotension, and depressed Glasgow Coma Scale scores [6, 7]. Injury patterns and the timeliness and quality of resuscitative and surgical interventions further influence prognosis, underscoring the importance of early recognition and coordinated trauma care [6, 8].

We report a 37-year-old male referred from a peripheral facility in Nakasongola District, Uganda, with an open chest injury occurring in the context of severe polytrauma following an occupational machinery accident. This case report has been reported in line with the SCARE criteria (Surgical CAse REport) 2025 guidelines, ensuring comprehensive and transparent reporting of the clinical details [9].

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2 | Case Presentation/Examination

A 37-year-old right-handed male machine operator was referred from a peripheral health facility, approximately 60 km northwest of Uganda's Capital, to our tertiary center in Kampala following a workplace accident in which a faulty wood-processing machine struck the posterolateral aspect of his right chest. The patient sustained a large open chest injury with visible protrusion of lung tissue, profuse bleeding from the wound, and immediate inability to turn his neck to the right side. He had no known past medical or surgical history, no documented history of regular medication use or known drug allergies, and no significant family history of inherited disease, and he was previously independent in activities of daily living. He was the primary breadwinner for a household of seven children. Initial care including analgesia was provided at the referring facility before transfer, where he arrived approximately 11 h after injury. There was a delay in accessing an ambulance. He was escorted by a nurse in an ambulance and received supplemental oxygen via nasal prongs during transport. The patient had no cervical collar in place on arrival. Apart from severe breathing difficulty and chest pain, he reported no other systemic symptoms on arrival.

On examination, the patient was in severe respiratory distress with a patent airway and marked use of accessory muscles of respiration. He had severe pallor. There was external herniation of the lung with respiratory movements (Figure 1). Chest expansion was asymmetric (left greater than right) with significantly reduced air entry on the right side. Respiratory examination revealed normal left-sided breath sounds but absent breath sounds and dullness to percussion over the right hemithorax, associated with palpable subcutaneous crepitus and paradoxical sucking movement of the chest wall. Once fully exposed, a massive gaping right-sided posterolateral wound and an additional laceration were seen in the right axilla. He was receiving supplemental oxygen via nasal prongs at 5 L/min. The peripheries were warm with capillary refill time of less than 3 s. Neurologically, he was alert and oriented with a Glasgow Coma Scale score of 15, and pupils were equal and reactive to light. At presentation, vital signs showed a heart rate of 88 beats per minute, oxygen saturation of 95% on 5 L/min of oxygen, and blood pressure of 104/76 mmHg.

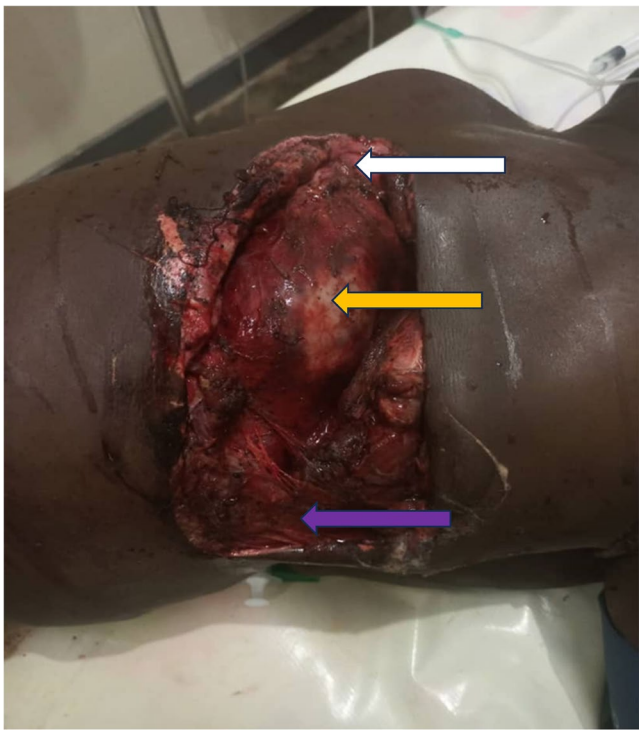


FIGURE 1 | Photograph showing the posterior chest wall wound at admission. The image demonstrates herniated lung tissue with an area of exposed lung lacking parietal pleural covering (white), parietal pleura (yellow), and surrounding crushed chest wall musculature (purple).

3 | Methods (Differential Diagnosis, Investigations and Treatment)

An Extended Focused Assessment with Sonography for Trauma (E- FAST) demonstrated free fluid in the right pleural space, consistent with a hemothorax, with no pericardial or intra-abdominal free fluid. Given the presence of an open chest wound with lung evisceration and severe respiratory distress, immediate surgical exploration under anesthesia was planned. Blood samples were obtained for grouping and crossmatching, and two units of blood were requested and secured.

Initial resuscitation prioritized airway, oxygenation, hemorrhage control, and circulation. The open chest wound was covered with sterile gauze and a three-sided occlusive dressing to control bleeding, prevent further lung evisceration, and limit air entry while allowing exit of intrathoracic air. Initial circulatory support with intravenous crystalloids and tranexamic acid was administered while awaiting blood availability, alongside intravenous paracetamol and fentanyl for analgesia and ceftriaxone for broad-spectrum antibiotic prophylaxis. Oxygen therapy was escalated to 10 L/min via a non-rebreather mask as his oxygen saturation was less than 92% on nasal prongs. A cervical collar was placed due to the high-energy mechanism of injury and the presence of neck pain with restricted range of motion, as cervical spine injury could not be excluded at presentation. Informed consent for emergency surgical intervention, including wound closure and chest tube insertion, was obtained from the next of kin, and the on-call surgical and anesthesia teams were notified as preparations were made for emergency theater.

After discussion with the on-call consultant, two senior surgery residents wheeled the patient into the emergency theater. Under general anesthesia and aseptic technique, surgical exploration of the wound revealed a 30-cm laceration extending from the posterior chest wall to the mid-axillary line, spanning the fifth and sixth intercostal spaces, as well as an additional 10-cm laceration in the axilla. Multiple puncture sites were identified in the parietal pleura, the largest measuring approximately 5 cm, along with numerous fractured ribs, bone fragments, and a peripheral diaphragmatic tear. A large volume of intrathoracic blood clots was evacuated. The diaphragmatic tear was repaired with Nylon 1 sutures, the parietal pleural defects were closed with Vicryl 2/0 sutures, and the intercostal muscles were reapproximated with Vicryl 1, with the approximation of the remaining back muscles using Vicryl 3/0. An intrathoracic chest tube drain was inserted and secured at 20 cm, after which the subcutaneous layers were closed and dressed in iodine-impregnated gauze (Figure 2).



FIGURE 2 | An intraoperative image showing the axillary wound after it was repaired.

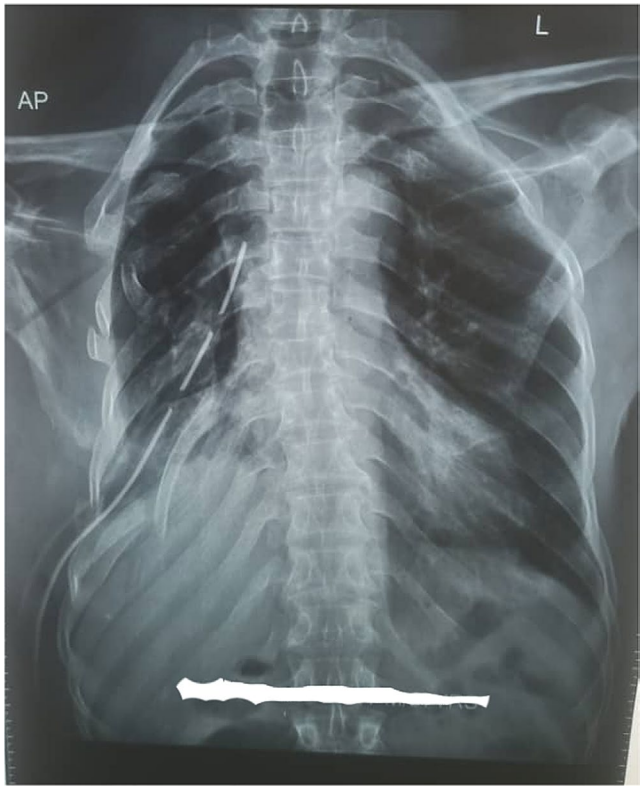


FIGURE 3 | A chest radiograph showing a chest tube in situ and multiple fractures of the ribs and scapula.

Postoperatively, the patient was admitted to the high-dependency unit for close monitoring. He was maintained on oxygen via a non-rebreather mask at 10 L/min and received intravenous fluids. Analgesia followed the WHO step-3 analgesic ladder and included a multimodal regimen of intravenous paracetamol, diclofenac, and a fentanyl infusion. Broad-spectrum intravenous antibiotics (ceftriaxone and metronidazole) were continued. Additional supportive measures included two-hourly vital sign monitoring, tetanus toxoid administration, stress ulcer prophylaxis with intravenous rabeprazole, and transfusion of two units of packed red blood cells within the first 24 h following surgery, as he had pallor.

4 | Outcome and Follow-Up

A tertiary survey was performed approximately 12 h after surgery. The airway remained patent. Breathing assessment revealed a functioning right-sided chest drain with approximately 400 mL of bloody output, while oxygen saturation ranged between 88% and 92% on supplemental oxygen. Circulatory parameters showed a blood pressure of 104/76 mmHg and a pulse rate of 113 beats per minute. Neurologically, the patient remained at GCS of 15; however, marked weakness of the right upper limb was noted, with associated edema and muscle power graded at 1/5. Distal pulses in the right arm were diminished compared to the contralateral side. On exposure, the surgical dressing was found to be soiled, with clear fluid draining from the posterior wound site. Further systemic examination revealed no scalp or facial injuries, no epistaxis or otorrhea, and intact facial bones. Neck examination demonstrated mild right-sided tenderness with restricted range of motion. The abdomen was soft, non-distended, and non-tender, with no guarding, and the pelvis was clinically stable. Urine output over the preceding 12 h was approximately 500 mL of clear urine without hematuria. Examination of the extremities demonstrated swelling and bruising of the right upper limb, with reduced peripheral pulses on the affected side; pulses in the remaining limbs were normal.

In view of the evolving respiratory, neurological, and vascular findings, oxygen therapy was escalated to 15 L/min via a non-rebreather mask. A chest radiograph was obtained, followed by computed tomography of the chest, cervical spine, thoracic spine, and head, which revealed multiple right-sided rib fractures, a right scapular fracture, and a sternal fracture, associated with a right-sided pneumothorax and partial collapse of the right lung (Figures 3 and 4). Additional findings included pneumomediastinum, bilateral chest wall and neck emphysema, air within the cervical and thoracic spinal canals, pneumocranium, and bilateral pleural effusions (smaller on the left). The chest tube was noted to be malpositioned (Figure 5). The initial chest tube was therefore removed and replaced with a 32-French thoracostomy tube, after which repeat imaging was planned. Neurosurgery and orthopedic teams were consulted.

After air was identified in the thoracic spinal canal, the CT scan of the head and spine performed to assess for intracranial or spinal injury revealed degenerative changes but no acute cervical or thoracic spinal fractures and appropriate positioning of the newly inserted thoracostomy tube. There was air in the cervical and thoracic spinal canals and extra-axial pneumocranium. Ongoing supportive care included incentive spirometry,

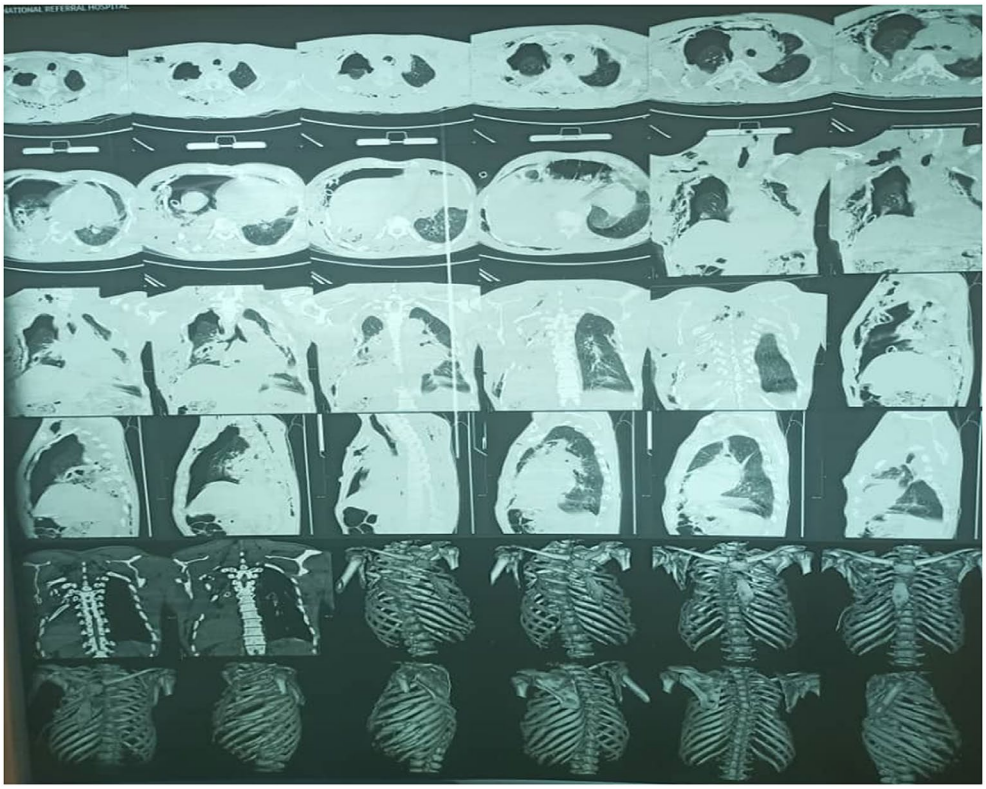


FIGURE 4 | An image of the chest CT scan showing multiple rib fractures and partial lung collapse.



FIGURE 5 | 3D reconstruction of chest wall deformity showing kinking of the chest tube (yellow arrow showing chest tube outline, while blue mark shows kinked area).

ambulation as tolerated, high-protein nutritional support and soft diet, daily wound dressing, and routine chest tube care. Pharmacologic thromboprophylaxis with low-molecular-weight heparin was initiated.

Laboratory investigations on postoperative day 1 revealed leukocytosis with a white blood cell count of $20\times10^3/\mu L$, hemoglobin of 15.5 g/dL, and platelet count of $297\times10^3/\mu L$. By postoperative day 3, the white blood cell count had decreased to $9.98\times10^3/\mu L$, hemoglobin was 9.9 g/dL, and platelet count was $226\times10^3/\mu L$. This drop was possibly dilutional. Liver function tests demonstrated hypoalbuminemia with a serum albumin level of 2.5 g/dL, while inflammatory markers were elevated, with a C-reactive protein level of 187 mg/L. Renal function tests remained largely within normal limits throughout the postoperative period.

Neurosurgery recommendations for the presence of pneumocranium on CT scan (Figure 6), without tension physiology, included maintaining bed rest in the 30-degree Fowler's position and on high-flow oxygen. Furthermore, he received intravenous ceftriaxone 2 g twice daily, and intravenous metronidazole for 1 week as well as pneumococcal vaccination, intravenous proton pump inhibitor (esomeprazole), and analgesia consisting of paracetamol and a fentanyl patch. He underwent daily chest physiotherapy with incentive spirometry and daily wound dressing changes and chest tube care. The chest tube output progressively reduced, as shown in Figure 7 below.

The patient had persistent right upper limb paralysis for which the orthopedic team was consulted and recommended

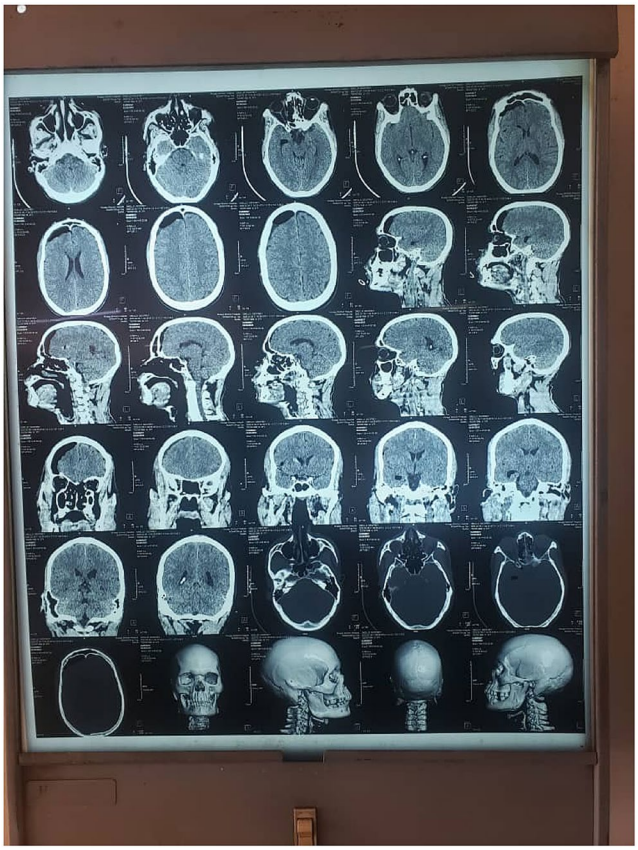


FIGURE 6 | Showing the patient's CT scan taken on post-operative day three, demonstrating pneumocranium.

management with a brachial plexus palsy sling. Over the following days, the patient remained clinically stable on the above treatment. Chest tube output continued to reduce, and the chest tube was removed on day 15th postoperative day (Figure 7).

Approximately 2 weeks later, the patient developed wound discharge. Wound culture and sensitivity testing grew isolated Achromobacter species, which is resistant to ceftriaxone and metronidazole but sensitive to piperacillin–tazobactam; antibiotics were therefore changed accordingly, and wound dressing was continued. Over the course of the third postoperative week, the patient developed swelling of the right arm (Figure 8). Doppler ultrasonography revealed right upper limb deep venous thrombosis involving the basilic and brachial veins. The patient was started on therapeutic low molecular weight heparin (enoxaparin) for 5 days and transitioned to apixaban.

Thirty-eight days after injury, the patient was discharged with referral to physiotherapy for stretching and strengthening exercises, warm compresses, as well as review by a hematologist. Apixaban was continued for a further 3 months with monthly hematologist reviews for the traumatic thrombosis in the right upper limb.

Six months following the injury, the patient complained of persistent weakness of the right upper limb and discomfort along the right chest. On examination, there was an obvious deformity of the right chest, with healed surgical scars (Figure 9). He still had muscle power of 1/5 in the right upper limb, with associated muscle atrophy, as shown in the image. Apixaban was stopped

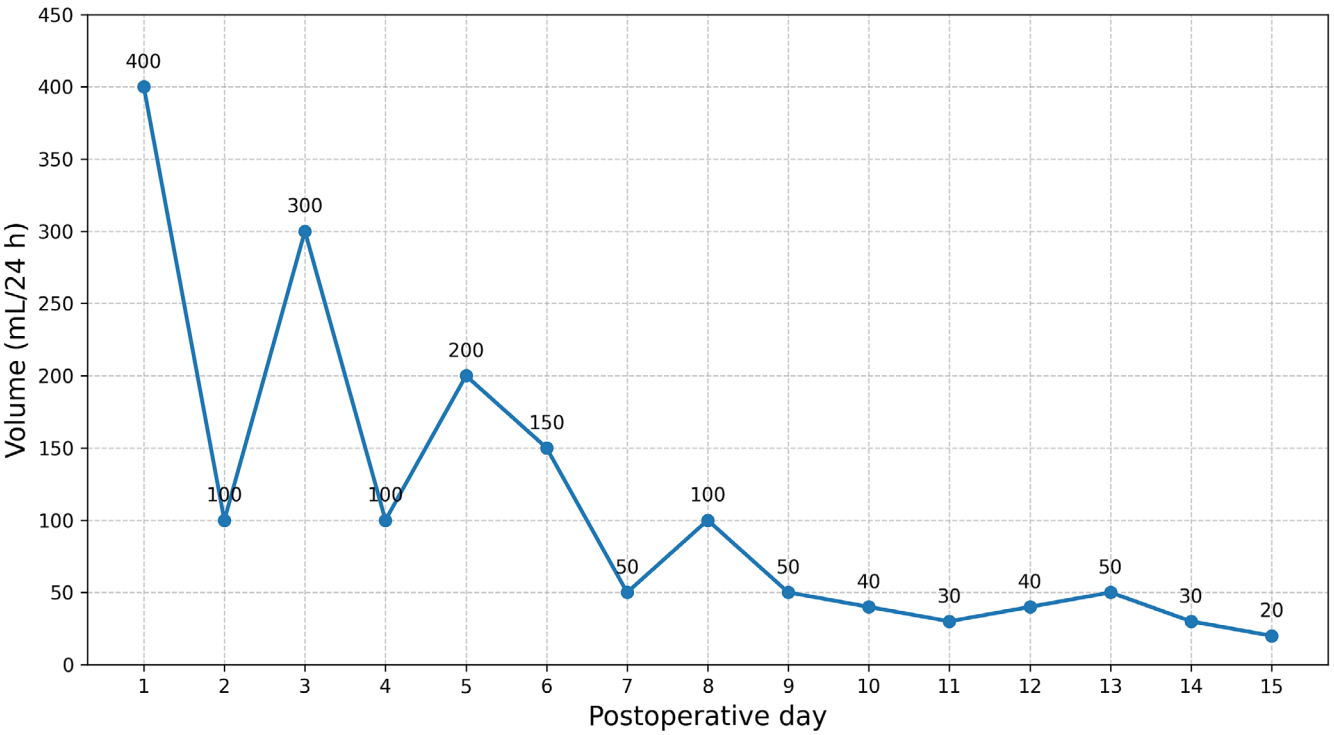


FIGURE 7 | Trend in daily chest tube output during the postoperative period.



FIGURE 8 | Right upper limb swelling caused by thrombosis in basilic and brachial veins.



FIGURE 9 | At 6 months, showing reduction in size of the right upper limb compared to left limb. There was visible irregularity at the scar site on the right side of the chest.

by the hematologist, and he was advised to continue physiotherapy. He reported that the prolonged hospitalization and persistent functional impairment resulted in loss of income, as the patient was unable to resume his occupation as a machine operator, placing a substantial financial burden on the patient and his dependents during recovery.

5 | Discussion

This case illustrates the extraordinary complexity of combined penetrating and blunt chest trauma with lung evisceration, complicated by rare sequelae including pneumocranium, brachial plexus injury, and upper extremity deep vein thrombosis. The delayed presentation and subsequent management highlight both the challenges and adaptive strategies required in resource-limited trauma systems.

Chest trauma with lung evisceration is typically the result of high-energy mechanisms, as demonstrated in this case [10, 11]. Such injuries are frequently associated with intrathoracic and extra-thoracic injuries [12], as evident in our patient including: pneumomediastinum, peripheral diaphragm tears, parietal pleural tears, scapular and sternal fractures, cervical and chest wall emphysema, and the presence of air within the cervical and thoracic spinal canals as well as the cranial cavity (pneumocranium). The severe chest injury allowed air to dissect along fascial planes via the Macklin effect, leading to pneumomediastinum and extension of emphysema into surrounding anatomical regions [13, 14].

Pneumocranium is an unusual finding in thoracic trauma. In this patient, air likely entered the cranial cavity through a dural defect or via negative intracranial pressure (the inverted soda bottle or suction effect) following cerebrospinal fluid leakage [15, 16]. This latter mechanism is supported in our case by the observation of clear fluid leaking from the posterior wound on the first postoperative day, which may have represented CSF. Despite the presence of intracranial air, he remained neurologically intact, and conservative management like bed rest, high-flow oxygen, and antibiotics was sufficient [17]. This underscores the importance of recognizing rare complications early and tailoring treatment according to clinical indications rather than defaulting to invasive intervention based on imaging findings alone.

The brachial plexus injury likely resulted from a combination of traction and stretch during the high-energy crush mechanism, direct contusion from the axillary laceration, and later compressive effects from hematoma and venous thrombosis. Such combined blunt-penetrating thoracic injuries can produce diverse neurovascular pathology, as illustrated here. Diagnosis was based on clinical assessment, serial motor grading, sensory testing, and distal pulse evaluation, since advanced studies such as nerve conduction, electromyography, or MRI were unavailable. The proximity of the axillary wound to the plexus and absence of cervical spine fracture supported the diagnosis, though precise localization and prognosis would ideally require electrophysiological or imaging confirmation, which was a limitation in this case.

Management followed ATLS principles, prioritizing immediately life-threatening injuries [18]. Initial stabilization was achieved using high-flow oxygen via a non-rebreather mask without immediate endotracheal intubation. While delaying intubation until arrival in the operating room may be appropriate in selected stable patients, early intubation should be considered when delays in transport or access to the operating room are anticipated, as clinical respiratory deterioration may

occur rapidly. Regarding diagnostic evaluation, an E-FAST provides a quick assessment for potential life-threatening injuries [19] to facilitate triage of interventions indicated. Although chest CT would provide superior anatomical detail for definitive operative planning, use must be balanced against the risk of delaying surgical management in unstable patients. In this case, priority was placed on rapid resuscitation and transfer to the operating room, with detailed CT imaging deferred to the postoperative period to assess for further injuries [20]. Where available, early postoperative CT angiography would have been a valuable adjunct to evaluate for associated subclavian, axillary, intercostal, or other thoracic vascular injuries, particularly given the reduced distal pulses in the affected upper limb. In this case, CT angiography was not obtained because of financial and logistical constraints, highlighting an important systems-level limitation in resource-constrained trauma care [20].

In open chest injuries with lung evisceration, initial management involves maintaining the lung tissue in situ and covering it with saline-moistened gauze [10]. This creates a temporary seal over the chest wall defect, helps prevent progression to a massive pneumothorax, and may assist in tamponading hemorrhage which was done in our case. Reduction of the lung is therefore deferred until controlled conditions in the operating room [10]. In severe thoracic trauma with major hemorrhage or physiological instability, damage control surgery prioritizes rapid bleeding and contamination control, temporary chest closure with chest tube drainage, and planned re-exploration after stabilization [21]. Although damage control was considered, the patient remained hemodynamically stable without evidence of profound physiological derangement, allowing definitive repair to be safely completed during the initial operation.

The clinical course was further complicated by systemic barriers, including an 11-h delay before presentation, ambulance transport challenges, and limited access to advanced critical care [22, 23]. These factors shaped management priorities, with rapid resuscitation and surgery taking precedence over immediate advanced imaging. Despite these constraints, multidisciplinary input proved vital. Neurosurgery guided conservative management of pneumocranium, orthopedics addressed brachial plexus palsy, and hematology managed venous thrombosis. While functional recovery was incomplete, coordinated care reduced morbidity and prevented mortality. The patient's prolonged hospitalization, need for anticoagulation, and permanent disability also highlight the socioeconomic burden of occupational trauma, particularly in contexts where financial protection mechanisms are limited [24, 25].

This case highlights the complexity of managing severe crush penetrating chest trauma with lung evisceration and rare associated complications in a resource-limited setting. Early adherence to ATLS principles, timely surgical intervention, and coordinated multidisciplinary team involvement were critical to survival and subsequent complication management. Despite successful acute care, the long-term functional impairment underscores the need for strengthened trauma systems, rehabilitation services, occupational safety measures, and financial protection mechanisms to mitigate the broader socioeconomic impact of trauma.

Author Contributions

Brian Kasagga: conceptualization, data curation, investigation, methodology, project administration, resources, visualization, writing – original draft, writing – review and editing. **Jamal-diin Lubega Wamala:** visualization, writing – original draft, writing – review and editing. **Dan Otim:** visualization, writing – review and editing. **Ketra Venessa Nandera:** writing – review and editing. **Godfrey Kikuba:** investigation, resources, writing – review and editing. **Dan Ssekiwunga:** resources, visualization, writing – review and editing. **Paul Ssempebwa:** writing – review and editing. **John Paul Magala:** supervision, writing – review and editing. **Emmanuel Elobu:** supervision, writing – review and editing. **Wilberforce Kabweru:** supervision, writing – review and editing. **Rachel Koch:** conceptualization, methodology, supervision, validation, writing – review and editing.

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Ethics Statement

As a single case report with the patient's signed consent, no additional ethical approval was required.

Consent

Written informed consent was obtained from the patient for publication of this case report and any accompanying images. A copy of this was provided to the editor during submission.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

For this case report, clinical information was obtained from patient files. Additional details are available upon reasonable request, subject to compliance with hospital privacy regulations.

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